SOUTHWIRE · POWER GENERATION SOLUTIONS

BUILT TO PULL.

Cable installation solutions for power generation, from reel to energised.

FOR EPCS AND ELECTRICAL CONTRACTORS

SIMPULL · MAXIS · PREFAB · CONTRACTOR SOLUTIONS

Stephan Hardt · stephan.hardt@southwire.com

EVERY CIRCUIT. EVERY PULL. ON SCHEDULE.

A 1,700 MW combined-cycle campus with carbon capture, battery storage and modular power is more than five million feet of cable, nearly six thousand circuits, and hundreds of pulls that all have to happen in the right order, in outage windows, around live plant.

This brochure follows that cable through a project, from the schedule to the energised plant, and then walks the campus zone by zone: what makes installation different there, and what Southwire brings to make it safer, faster and cleaner. Every image and number comes from one 3D model of the plant and its cable schedule.

5,687
CIRCUITS

19,715
CABLES

5.33 M
FEET OF CABLE

1,468
REELS (EST.)

13
VOLTAGE CLASSES

SK-3X1 campus at golden hour: three H-class GTs, HRSGs and ST, ACC, 230 kV switchyard, carbon capture, BESS and modular power.

FROM REEL TO ENERGISED

01 PLAN
Model, route and size every circuit; plan pulls and reels

02 DELIVER
Cut-to-length, paralleled, colour-coded reels to the pull

03 STAGE
One person moves the reel; no jacks, no pallet jacks

04 PREFAB
Spine modules, pre-terminated MV, skids pre-wired

05 PULL
No lube, factory heads, safer pullers

06 TERMINATE & TEST
Terminations and test records for turnover

07 OPERATE
Outage support, training, people on site

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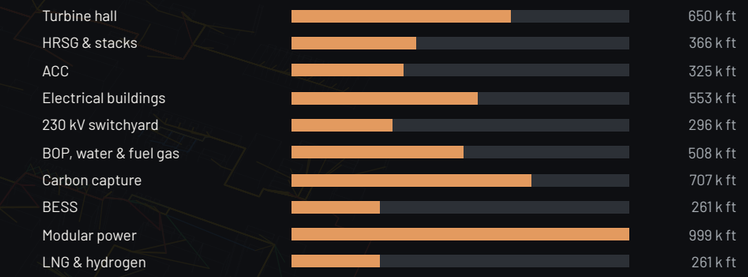
02 — PLAN

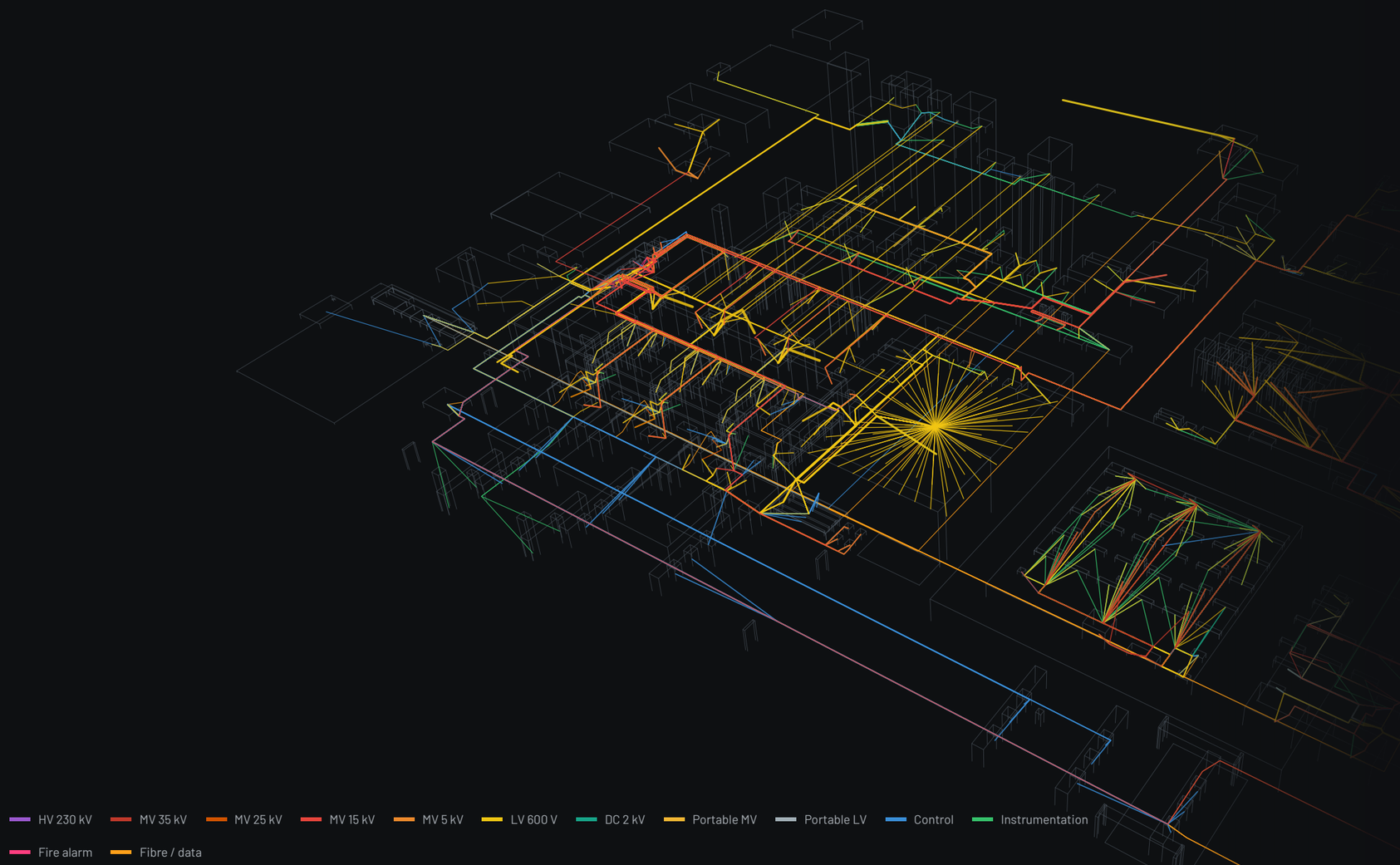
SEE EVERY CABLE BEFORE YOU PULL IT.

Pull planning starts long before the first reel ships. On SK-3X1 every circuit is in the model with its route, length, size and voltage-drop check, so reels can be cut to the pull and the pull sequence planned.

- **CONFIGURATOR PLUS WEB APP** plan, manage and order feeder reels from any device
- **SIMPULL CABLE PULL CALCULATOR** conduit size and fill, pulling tension, sidewall pressure, jam probability, voltage drop, ground sizing
- **VOLTAGE DROP AND CONDUIT FILL CALCULATORS** size conductors and conduit to NEC from the field
- **CONTRACTOR SOLUTIONS PROFESSIONALS** on site to consult on layouts, pulling calculations and set-up

ROUTED CABLE BY ZONE





Cable X-ray of the SK-3X1 model: 908 routed paths carrying the home-run cables of the schedule, coloured by voltage class and weighted by cable count. Device tails to local junction boxes are counted, not drawn.

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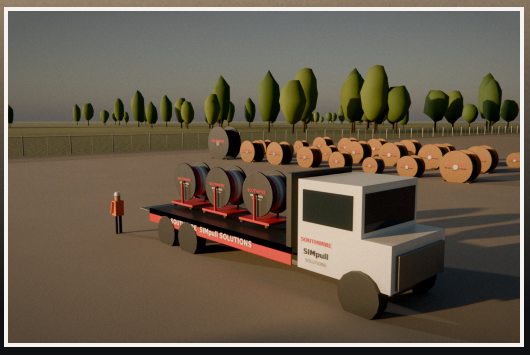
Modular yard: the SIMpull Truck brings the feeder reels to the 13.8 kV cable pull.



Reel yard: SIMpull Reels coming off the truck.

THE RIGHT REEL, AT THE RIGHT PULL.

Reels arrive cut to the length of the pull, paralleled and colour-coded, labelled for the location they feed, and on payoffs that roll straight to the conduit. On SK-3X1 the SIMpull Truck brings the 13.8 kV collector replacement reels to the modular-yard pull.

- **SIMPLE DISTRIBUTION** five fully stocked customer service centers across the US
- **SIMPLE DELIVERY** colour, paralleled, cut-to-length reels with SIMpull Heads, delivered on SIMpull payoffs on the SIMpull Truck
- **SIMPULL STACK REELS** one reel, up to five coloured conductors per pull and up to four pulls, shrink-wrapped between layers
- **LABELLED REELS** each reel tagged for its location, so the job site knows where it goes
- **RIP CHIP** removes the protective wrap without knives
- **FEWER REELS ON THE JOB** less scrap and clean-up, fewer reels to move and set up

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The RR-SW45 Reel Roller: a 35 kV MV-105 steel reel on driven rollers, pendant in reach, operator out of the bite.

Reel yard staging on the SK-3X1 laydown pad.

ONE PERSON. NO JACKS.

Set-up used to mean reel jacks, axles, pallet jacks and a crew. Staged on SK-3X1's reel yard: SIMpull Reels on Cradles, 45K and 30K Reel Rollers carrying the MV steel reels, SIMpull Flanges on wooden reels and Stack reels on JAX stands, with the temporary power and lighting to run the yard.

- **SIMPULL REEL** 61 x 32 in, up to 6,000 lb, moved by one person, fits through 36 in doors
- **SIMPULL CRADLE** holds the 61 in reel, pays off directly, forklift and pallet-truck manoeuvrable
- **SIMPULL FLANGE** shaftless, field-installable on wooden or plastic reels up to 34 in and 2,000 lb
- **RR-SW45 / RR-SW30 REEL ROLLERS** 45,000 / 30,000 lb, VFD pendant from outside the wire path, stackable
- **MAXIS JAX / PRO-JAX** reel stands that lift reels safely; PRO-JAX to 10,000 lb
- **TEMPORARY POWER & LIGHTING** generator to cam-lock cords, 480 V carts, distribution boxes, work lights

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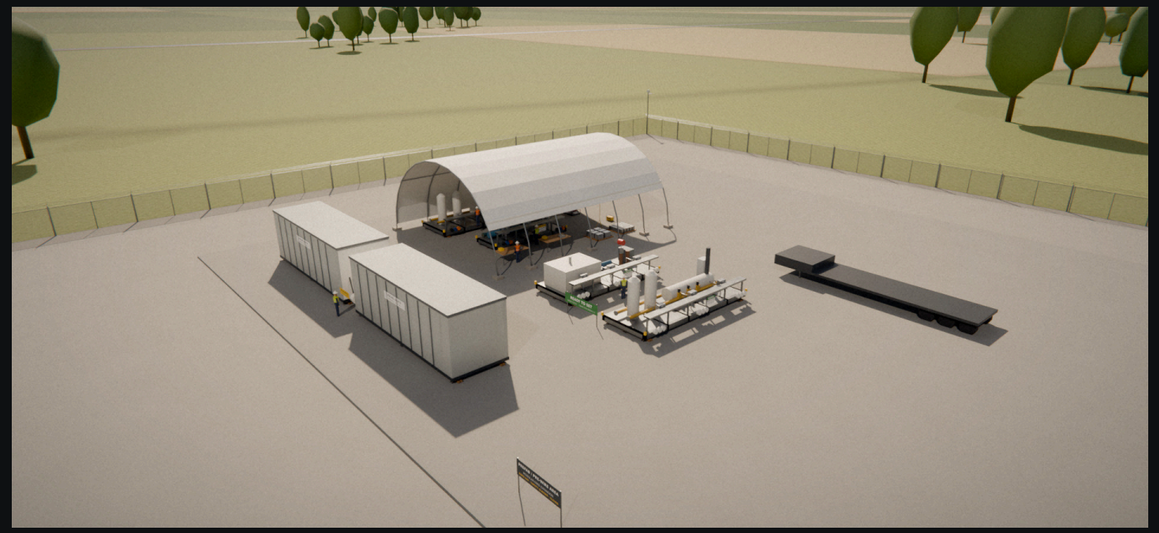
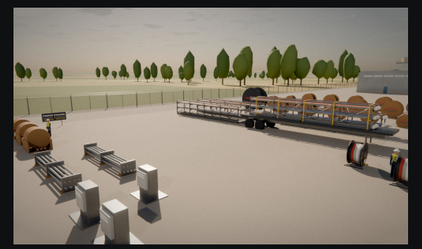
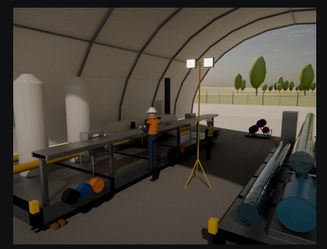
PULL IT ONCE. AT THE SHOP.

Modular-yard expansion: spine module 5 of 9 comes down on its bents, modules 1-4 already set from the MOD-EH e-house; electricians on the boom lift take the coiled tails.



Skid pre-wire workstation in the maintenance laydown, away from energised plant: skids wired under cover, tested skids ready to set, a PDC e-house in shipping splits.





Every hour of cable work moved off the critical path and into a shop or a laydown is an hour that does not happen at height, in an outage window, or next to live equipment.

- **CABLE SPINE MODULES** 40 ft frames with three ladder-tray tiers, cable laid and tied at the shop, coiled tails at the joint; set, spliced, done
- **PRE-TERMINATED MV ASSEMBLIES** cut to length, factory terminations, tested and capped before they ship
- **SKID PRE-WIRE** JBs, skid-edge tray, conduit and tagged tails at waist height under cover; IR and continuity recorded before the pick
- **SHIPPING-SPLIT INTERCONNECTS** PDC e-house splits arrive with pre-terminated interconnects coiled at the split
- **PREFAB CONDUIT RACKS** multi-conduit bundles on frames, set like structure

Prefabricated cable systems are shown generically; scope and product names to be confirmed with Southwire.

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Factory SIMpull Heads on paralleled SIMpull THHN going into the wireway: no lube, no field-made pulling head.

NO LUBE. FACTORY HEADS. FEWER HANDS.

On SK-3X1 the welding-shop feeder goes in as five paralleled SIMpull THHN conductors off one SIMpull Reel, factory SIMpull Heads straight into the wireway, a tension monitor and a Triggers switch on the pull.

- **SIMPULL THHN, NOLUBE** installs without pulling lubricant into conduit; installation performance guaranteed
- **SIMPULL HEAD GRIPS** factory-installed, low-profile, staggered; about 3 ft of cable per phase not wasted
- **COILPAK AND BARREL** pay off straight from the package; BARREL up to 7 conductors for home runs
- **MAXIS XD10 / XD1 / FEEDER** up to 10,000 lb; drill-powered 600 lb constant for branch circuits
- **EXTREME DUTY TENSION MONITOR** 10,000 lb working load, USB data for the pull record
- **TRIGGERS WIRELESS SAFETY SWITCH** controls input power to job-site equipment

TRADITIONAL

Feeders the traditional way: 15 reels, 1 pallet jack, 5 reel jack stands, 5 axles, 7 electricians.

SIMPULL REEL PAYOFF

1 SIMpull Reel Payoff, 2 electricians. Example as published in the Southwire SIMpull Solutions brochure.

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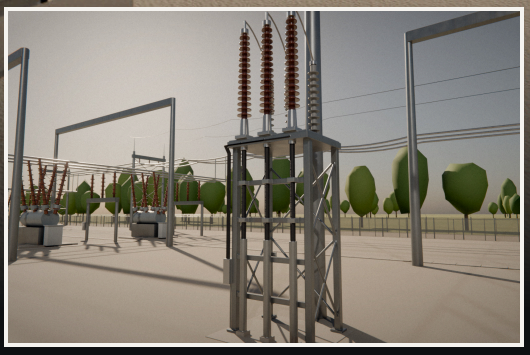
BESS PCS / MV skid: 34.5 kV elbow terminations to the collector circuit.



230 kV XLPE sealing ends at the switchyard.

TERMINATED, TESTED, RECORDED.

Turnover is a package of test records. Pre-terminated assemblies arrive tested; skid wiring is meggered at the bench before the pick; tension-monitor data documents each pull. In the field: MV elbows at the BESS skids, 230 kV sealing ends at the switchyard, bottom-entry terminations under the switchgear.

- **PRE-TERMINATED MV** factory terminations, tested and capped; fewer field terminations at height
- **BENCH TESTING** insulation resistance and continuity recorded at the pre-wire workstation
- **PULL RECORDS** tension monitor logs with each pull, ready for the turnover file
- **PRINTED LEGENDS** every jacket identified in the tray for QA walk-downs
- **ARMOR-X MC-HL** armored, jacketed cable for trays, racks and hazardous areas
- **230 KV TERMINATIONS** cleated risers, sealing ends, arresters, link boxes

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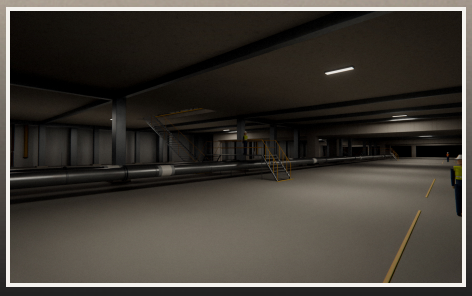
GT1 outage: a new 13.8 kV feeder off the RR-SW45 in the laydown bay, up into the EL 48 MV tray.



Under the deck: stairs and stiles between units.

TURBINE HALL

721	3,329	650 k	1,500
CIRCUITS	CABLES	FT ROUTED	FT LONGEST RUN

Instrumentation ITC (IS)	163 k
Instrumentation PLTC / TC-ER	148 k
Control TC-ER #14	84 k
ARMOR-X MC-HL control	68 k

WHAT'S DIFFERENT HERE

Outage windows and a crowded hall. MV feeders climb from the laydown bay to trays at EL 41-49 over the machines; reels are big, floor space is small, and every pull competes with the outage critical path.

THE INSTALLATION SOLUTION

- **RR-SW45 IN THE LAYDOWN BAY** turns the steel reel from the pendant, operator out of the bite
- **CUT-TO-LENGTH OUTAGE REELS** sized and labelled per pull, staged before the unit trips
- **MAXIS XD10 + TENSION MONITOR** at the tray drop, pull recorded for turnover
- **ARMOR-X AND MV-105 WITH LEGENDS** tray identification for QA walk-downs

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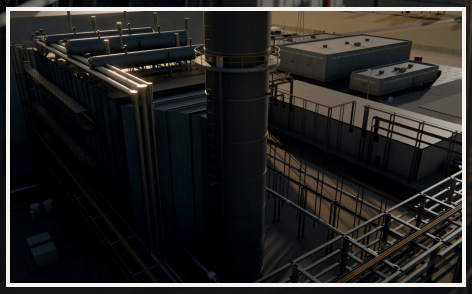
HRSG 1 west wall: buckstays, SCR doors, downcomers, blowdown tank.



Stack platforms and the CEMS shelter.

HRSG & STACKS

582
CIRCUITS

1,716
CABLES

366 k
FT ROUTED

1,450
FT LONGEST RUN

Instrumentation PLTC / TC-ER	203 k
Thermocouple ext. KX	59 k
Control TC-ER #14	50 k
Type TC-ER 600 V power	45 k

WHAT'S DIFFERENT HERE

The densest instrumentation on the site: drum levels, thermocouples, CEMS and valve positions on platforms up the casing, many short runs to junction boxes, then multi-pair home runs to the electrical buildings.

THE INSTALLATION SOLUTION

- **CUT-TO-LENGTH INSTRUMENT REELS** labelled per JB, no wasted tails on the platforms
- **PRE-WIRED JB SKIDS** terminated and checked at the bench, set on the platform
- **MAXIS XD1** drill-powered pulls for the short conduit runs at height
- **PREFAB CONDUIT RACKS** platform conduit set like structure

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ACC south-east corner: street risers, stair tower, fan bells under the deck.

R4: 92 fan VFDs, output trays overhead.

ACC

419 CIRCUITS

881 CABLES

325 k FT ROUTED

1,630 FT LONGEST RUN

Instrumentation PLTC / TC-ER	118 k
Control TC-ER #14	86 k
VFD cable 600 V	64 k
Type TC-ER 600 V power	29 k

WHAT'S DIFFERENT HERE

Repetition: 80 fan cells and 12 aux coolers, each with a VFD feeder, control and vibration cables along the fan-bridge ladders. Identical pulls reward identical reels.

THE INSTALLATION SOLUTION

- **STACK AND PARALLELED REELS** one reel per street, each pull cut to the cell
- **VFD CABLE** symmetrical grounds and shield for the drive outputs
- **REEL LABELS BY CELL** the reel tells the crew where it goes
- **REEL ROLLERS ON THE DECK ACCESS ROAD** staged at the riser, no jacks

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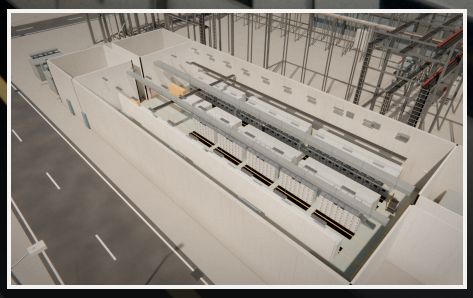
R1: 13.8 kV and 4.16 kV metal-clad lineups across the operating aisle.



R1 from above, roof and wall off.

ELECTRICAL BUILDINGS

621	1,690	553 k	1,550
CIRCUITS	CABLES	FT ROUTED	FT LONGEST RUN

Instrumentation PLTC / TC-ER	153 k
XHHW-2 1/C feeder sets	121 k
Control TC-ER #14	87 k
Type TC-ER 600 V power	64 k

WHAT'S DIFFERENT HERE

Where everything lands: bottom-entry MV terminations over the cable basement, paralleled LV feeder sets to the load centres and MCCs, and hundreds of control and instrument cables into a few rooms.

THE INSTALLATION SOLUTION

- **SIMPULL PARALLELED FEEDER SETS** colour-coded, cut to length, SIMpull Heads factory-fitted
- **PRE-TERMINATED MV** fewer terminations in the basement
- **NOLUBE XHHW-2 / THHN** into conduit without lubricant
- **COILPAK AND BARREL** lighting, receptacles and small power

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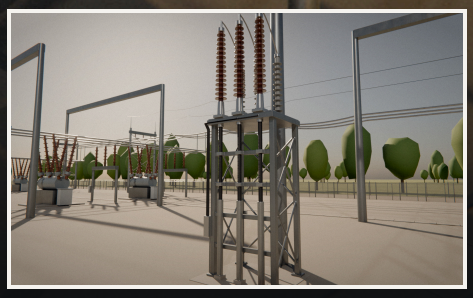
Bay D2: breakers, disconnects and the control-cable trench to the relay house.



230 kV XLPE sealing ends.

230 KV SWITCHYARD

214 CIRCUITS

586 CABLES

296 k FT ROUTED

1,950 FT LONGEST RUN

Control TC-ER #14	109 k
Instrumentation PLTC / TC-ER	100 k
Control TC-ER #10 (CT / VT)	82 k
Type TC-ER 600 V power	4 k

WHAT'S DIFFERENT HERE

Two worlds: 230 kV XLPE on heavy steel reels, and hundreds of control and CT / VT circuits in precast trenches from every breaker back to the relay house.

THE INSTALLATION SOLUTION

- **RR-SW45 FOR STEEL REELS** 45,000 lb, stackable, remote pendant
- **TENSION-MONITORED HV PULLS** sidewall pressure planned with the Cable Pull Calculator
- **CONTROL CABLE CUT PER BAY** reels labelled by diameter and breaker
- **CONTRACTOR SOLUTIONS PROFESSIONALS** pulling calculations and set-up on site

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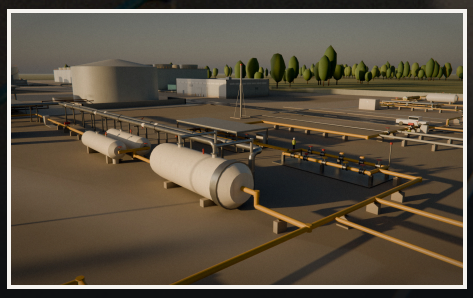
Tank farm: raw, fire / service and demin water, the fire pump house.



Plant gas yard: hazardous area.

BOP, WATER & FUEL GAS

455
CIRCUITS

1,414
CABLES

508 k
FT ROUTED

2,860
FT LONGEST RUN

Instrumentation PLTC / TC-ER	158 k
Type TC-ER 600 V power	143 k
Control TC-ER #14	81 k
ARMOR-X MC-HL 600 V	43 k

WHAT'S DIFFERENT HERE

Many small motor feeders, lighting and receptacles across a spread-out site; in the gas yard and M&R, Class I Div 2 areas with buried conduit and armored cable.

THE INSTALLATION SOLUTION

- **COILPAK, BARREL, WIRE WAGON** small power and lighting off the package, no broken spools
- **ARMOR-X MC-HL** hazardous-area power and control
- **SIMPULL NOLUBE IN BURIED CONDUIT** long underground runs without lube
- **TEMPORARY POWER AND LIGHTING** for the spread-out work fronts

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Train A: flue duct into the DCC, booster fan, overhead duct to the absorber.

Absorber A: tray riser and platform lights.

CARBON CAPTURE

751
CIRCUITS

2,753
CABLES

707 k
FT ROUTED

2,850
FT LONGEST RUN

Instrumentation PLTC / TC-ER	424 k
Type TC-ER 600 V power	101 k
Control TC-ER #14	67 k
MV-105 15 kV	36 k

WHAT'S DIFFERENT HERE

The largest new-build scope on the site: long rack runs up to 2,850 ft, a dense instrument count on absorbers and strippers, big blower and compressor drives. Built while the plant runs.

THE INSTALLATION SOLUTION

- **PREFAB CABLE SPINE ON THE CCS RACK** modules set with the steel, cable already laid
- **ARMOR-X ON RACKS** armored runs without conduit
- **REEL ROLLERS AND STACK REELS** staged by rack bay
- **PULL PLANNING** long-run tensions checked before the reels are cut

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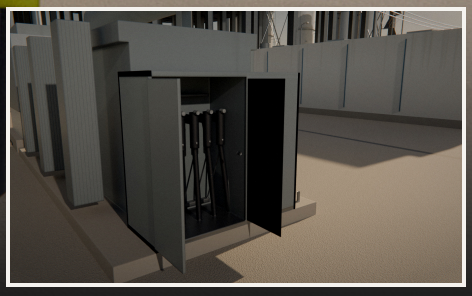
Crew pulling 1,500 V DC conductor into the precast trench.



PCS / MV skid elbows.

BESS

397	1,155	261 k	1,980
CIRCUITS	CABLES	FT ROUTED	FT LONGEST RUN

Instrumentation PLTC / TC-ER	86 k
PV / RHW-2 2 kV DC	59 k
MV-105 35 kV	40 k
Type TC-ER 600 V power	28 k

WHAT'S DIFFERENT HERE

Dozens of identical DC runs from containers to inverters, MV collector circuits to the e-house, and fast schedules.

THE INSTALLATION SOLUTION

- **CUT-TO-LENGTH PARALLELED DC REELS** one reel per block, labelled
- **SIMPULL HEAD GRIPS** pre-made, no field heads in the trench
- **MAXIS JAX STANDS** at the trench edge, no red jacks
- **PV / RHW-2 2 KV AND MV-105 35 KV** DC and collector cable

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MODULAR POWER.

Six technologies, three ways of cabling them. What changes is how long the power is meant to stay: permanent units get duct banks and prefab; bridge power goes on grade; factory skids arrive pre-wired.

1,091 CIRCUITS

4,846 CABLES

999 k FT ROUTED

5,610 FT LONGEST RUN

Instrumentation PLTC / TC-ER	509 k
Type TC-ER 600 V power	121 k
Control TC-ER #14	84 k
Thermocouple ext. KX	66 k

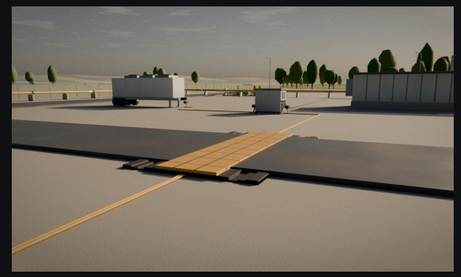
PERMANENT: RICE AND SIMPLE-CYCLE

8 RICE engines and two LM6000-class units: buried duct banks to the collector, skids pre-wired in the laydown, the prefab spine for LV / control / fibre.

791 CIRCUITS

715 k FT ROUTED

Prefab spine, skid pre-wire, SIMpull NoLube in duct banks



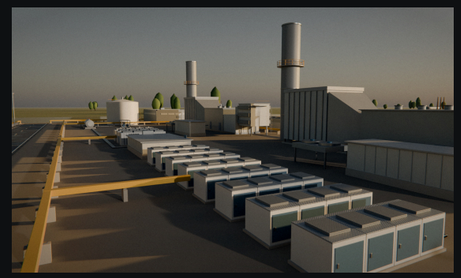
BRIDGE & TEMPORARY: TM2500, GENSETS, PORTABLE PAD

Power for months, not decades: cable on grade in ground trays, Type SHD-GC 15 kV and Type W with cam-locks, drive-over protectors.

167 CIRCUITS

189 k FT ROUTED

SHD-GC, Type W, cam-locks, temporary power & lighting



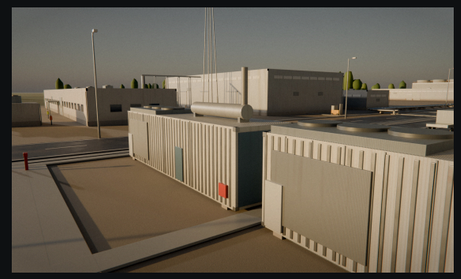
SMALL FACTORY SKIDS: FUEL CELLS, MICROTURBINES

Factory-built modules with short, repetitive connections: pre-wired, tested, set and connected.

112 CIRCUITS

74 k FT ROUTED

Pre-wired skids, CoilPAK / BARREL for small power



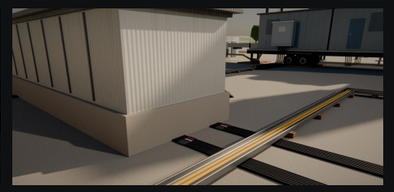
BLACK START & EMERGENCY GENSETS

Critical circuits that must work first time: tested and recorded.

21 CIRCUITS

20 k FT ROUTED

Tested assemblies, records for turnover



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LNG satellite: unloading arms, vaporizers, tanks in the impoundment.

Green hydrogen: electrolyzers and compressors.

LNG & HYDROGEN

384
CIRCUITS

1,245
CABLES

261 k
FT ROUTED

5,160
FT LONGEST RUN

Instrumentation ITC (IS)	140 k
MV-105 15 kV	40 k
ARMOR-X MC-HL control	38 k
ARMOR-X MC-HL 600 V	20 k

WHAT'S DIFFERENT HERE

Hazardous and cryogenic areas, intrinsically safe instrumentation, and the longest runs on the site, up to 5,160 ft to the send-out and electrolyzer buildings.

THE INSTALLATION SOLUTION

- **ARMOR-X MC-HL** hazardous-area power and control
- **ITC / PLTC (IS)** intrinsically safe instrument runs
- **LONG-LENGTH REELS ON ROLLERS** fewer splices on the long runs
- **CABLE PULL CALCULATOR** tensions and sidewall pressure for the long pulls

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Main control room: over the operator's shoulder, the plant on the video wall.

Six-position console and the 16-cube wall.

WE ARE WHERE YOU ARE.

The plant runs for decades and every outage brings cable back to the critical path. Southwire stays on the job: people on site, training for the crews, distribution close by, and a guarantee behind the cable.

- **CONTRACTOR SOLUTIONS PROFESSIONALS** on the jobsite for layouts, pulling calculations and set-up, with equipment to demo
- **SOUTHWIRE SOLUTIONS UNIVERSITY** hands-on training at the Thorn Customer Solutions Center, Carrollton, GA (15,000 sq ft)
- **SIMPULL PRODUCT PERFORMANCE GUARANTEE** SIMpull conductors install without lubricant as specified, or Southwire reimburses direct replacement costs (terms apply)
- **OUTAGE SUPPORT** cut-to-length reels, SIMpull Truck delivery and reel rollers staged ahead of the trip
- **SIMPLE DISTRIBUTION** five customer service centers
- **TOOLS & EQUIPMENT** Maxis pullers, Triggers, temporary power and lighting

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DON'T JUST PULL.
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